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Microwave Synthesis and Preliminary Antibacterial Activities of New Imidazolidines Derived from 2-Aminobenzothiazole

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ABSTRACT

2-aminobenzothiazole 1 was converted to the corresponding diazonium salt which was introduced in coupling reaction with alkaline solution of 2-hydroxybenzaldehyde as coupling reagent to give azo-benzothiazole derivative 2 bearing aldehyde group. The resulting aldehyde 2 was introduced in condensation reactions with the primary aromatic amines including (4-nitroaniline, 3-nitroaniline, 4-hydroxyaniline, 4-methoxyaniline, 2-methoxyaniline, 4-bromoaniline, 4-chloroaniline and 2,4-dichloroaniline) using microwave irradiation technique in absolute ethanol to produce eight imine derivatives of benzothiazole 3a-h, respectively. Treatment of the resulting imines 3a-h with L-alanine using microwave irradiation in tetrahydrofuran afforded eight new imidazolidines 4a-h substituted with benzothiazole moiety, respectively. Preliminary *in vitro* antibacterial activity of the target compounds were investigated using two types of bacteria, *Staphylococcus aurous* (Gram-positive) and *Escherichia coli* (Gram-negative). The results indicated that the newly synthesized imidazolidines (compounds 4c, 4d, 4f, 4g, and 4h) exhibited greater activities than gentamycin against Gram-positive bacteria. On the other hand, compounds 4c and 4f were also showed better activities against Gram-negative bacteria when compared with that of the control drug (Gentamycin).

Keywords: *Imidazolidines; Imines; Benzothiazoles; Azo; Antibacterial activity*,

INTRODUCTION

Imidazolidines (saturated imidazoles), also known as tetrahydroimidazoles are biologically active nitrogen containing heterocyclic moiety which have been reported to shown wide array of significant bioactivities[1]. Moreover, the construction of optically active stereogenic center in a saturated heterocycle such as imidazolidine[2]. Imidazolidines have found therapeutic applications in drugs such as the well established phenytoin3. Among the wide range of biological activities of substituted hydantoins we can cite antidiabetic anti-inflammatory, antiviral, GHS and CB1 receptor antagonists[3], and more. Imidazolidine ring of such compounds yields molecules with variety of pharmacological properties, including anti-inflammatory, antifungal, antibacterial, hypoglycemic, schistosomicides anti-cancer activities[4], antihypertensive/anaesthetic[5], antiarrhythmic[6], modulation of anxiety[7], antinociceptive activity[8], anti-Alzheimer's[9], antimicrobial[10], antinociceptive[11], antitumor[12], anti-Trypanosoma cruzi[13], anti- depressant[14] and antifungal drugs[15]. Benzothiazole consists of thiazole ring fused with benzene and is also called 1-thia-3-azaindene. It is a weak base and thermally stable molecule[16]. In the 1990s, various pharmacological investigations of newly synthesized benzothiazoles demonstrated interesting pharmacological activities and led to the development of new medications for treating diseases. They were studied extensively for their antiallergic, antiinflammatory[17], analgesic[18], anti-tumor activity[19], anti-proliferative activities[20]. Benzothiazoles have been reported to be antioxidant agents[21], antitubercular[22], antipsychotic[23], anticonvulsant[24], anticancer agents[25], anti-parasite activity[26] antimalaria[27].

Thus, in this article, we reported here the synthesis of new imidazolidine derivatives bearing the biologically active benzothiazole moiety which might have some biological activities.

2. EXPERIMENTAL

2.1. General

The chemicals were used as provided from Fluka, sigma aldrich and Merck. Microwave reactions were performed on Domestic microwave oven in crucible. Analytical TLC was performed with silica gel 60 F_{254} plates. The reactions were monitored by TLC and visualized by development of the TLC plates with iodine vapor. Melting points were recorded on an Electro thermal Stuart SMP 30 capillary melting point apparatus and are uncorrected. Infrared spectra were recorded on SHIMADZU FTIR–8400S Infrared Spectrophotometer as potassium bromide discs. ^{1}H NMR spectra was collected on Avance III 500, NMR spectrometer, Bruker, Germany at 500 MHz in DMSO–d_6 as solvent and TMS as an internal standard at a Faculty of Science, University of Tarbiat Modares, Iran. (CHNS) Elemental Analysis was carried out with Perkin Elmer 300A Elemental Analyzer at a Faculty of Science, University of Tarbiat Modares, Iran.

2.2. Chemical methods

2.2.1.**(*E*)-5-(benzo[d]thiazol-2-yldiazenyl)-2-hydroxy benzaldehyde (2)** was synthesized following the method described by Acton[28] as dark brown solid, mp 141-143 ^{o}C, yield 60 %; IR (cm^{-1}): 3429_{br} (νO-H), 3068 (νC-H, benzene), 2856 and 2715 (νC-H, aldehyde), 1645 (νC=O, aldehyde), 1610 (νC=N, benzothiazole), 1502 and 1475 (νC=C, benzene), 1375 (νN=N), 759 (δo.o.p.C-H, benzene).

2.2.2. **General procedure for the synthesis of imines 3a-h:**

All reactions were carried out on Domestic microwave oven in crucible. Reactions contained the azoaldehyde derivative **2** (0.283 g, 1 mmol), equimolar amount (1 mmol) of aniline derivatives (4-nitroaniline, 3-nitroaniline, 4-hydroxyaniline, 4-methoxyaniline, 2-methoxyaniline, 4-bromoaniline, 4-chloroaniline and 2,4-

dichloroaniline respectively) and absolute ethanol (1 mL). The crucible was introduced to the center of a Domestic microwave oven and then heated (300-350W) for 25-30 minutes. TLC (*n*-hexane: EtOAc) showed that the reactions were completed. The products were washed with diethyl ether and recrystallized from ethanol.

4-((E)-benzo[d]thiazol-2-yldiazenyl)-2-((E) -) ((4-nitrophenyl) imino) methyl)phenol (3a): IR (cm^{-1}): 3371 (νO-H), 3064 (νC-H, benzene), 1608 (νC=N, imine), 1589 (νC=N, benzothiazole), 1514 (νas.NO_2), 1448 (νC=C, benzene), 1404 (νN=N), 1336 (νs.NO_2), 758 (δo.o.p.C-H, benzene).

4-((E)-benzo[d]thiazol-2-yldiazenyl)-2-((E) -((3-nitrophenyl) imino) methyl)phenol (3b): IR (cm^{-1}): 3246 (νO-H), 3088 (νC-H, benzene), 1604 (νC=N, imine and νC=N, benzothiazole, vib. coupling), 1573 and 1502 and 1444 (νC=C, benzene), 1525 (νas.NO_2), 1396 (νN=N), 1354 (νs.NO_2), 758 (δo.o.p.C-H, benzene).

4-((E)-benzo[d]thiazol-2-yldiazenyl)-2-((E)- ((4-hydroxyphenyl)imino) methyl)phenol (3c): IR (cm^{-1}): 3369 and 3298 (νO-H), 3032 (νC-H, benzene), 1610 (νC=N, imine and νC=N, benzothiazole, vib. coupling), 1514 and 1460 (νC=C, benzene), 1379 (νN=N), 827 (δo.o.p.C-H, benzene).

4-((E)-benzo[d]thiazol-2-yldiazenyl)-2-((E) -((4-methoxyphenyl)imino) methyl)phenol (3d): IR (cm^{-1}): 3371 and 3298 (νO-H), 3037 (νC-H, benzene), 2949 (νas.CH_3), 1622 (νC=N, imine), 1595 (νC=N, benzothiazole), 1512 and 1456 (νC=C, benzene), 1392 (νN=N), 821 (δo.o.p.C-H, benzene).

4-((E)-benzo[d]thiazol-2-yldiazenyl)-2-((E)- ((2-methoxyphenyl)imino) methyl)phenol (3e): IR (cm^{-1}): 3462 and 3265 (νO-H), 3064 (νC-H, benzene), 2968 (νas.CH_3), 2870 (νs.CH_3), 1633 (νC=N, imine), 1608 (νC=N, benzothiazole), 1533, 1500 and 1444 (νC=C, benzene), 1363 (νN=N), 831 (δo.o.p.C-H, benzene).

E)-benzo[d]thiazol-2-yldiazenyl)-2-((E)- ((4-bromophenyl)imino)methyl)phenol

IR (cm^{-1}): 3369 and 3261 (νO-H), 3066 (νC-H, benzene), 1633 (νC=N, imine), 1608 (νC=N, benzothiazole), 1506 and 1489 (νC=C, benzene), 1363 (νN=N), 821 (δo.o.p.C-H, benzene).

4-((E)-benzo[d]thiazol-2-yldiazenyl)-2-((E)- ((4-chloropheny l)imino)methyl)phenol (3g)

: IR (cm^{-1}): 3369 and 3263 (νO-H), 3064 (νC-H, benzene), 1633 (νC=N, imine), 1606 (νC=N, benzothiazole), 1537, 1496 and 1442 (νC=C, benzene), 1367 (νN=N), 821 (δo.o.p.C-H, benzene).

4-((E)-benzo[d]thiazol-2-yldiazenyl)-2-((E)- ((2,4-dichlorophenyl)imino)methyl)phenol(3h): IR (cm^{-1}): IR (cm^{-1}): 3470 and 3265 (νO-H), 3061 (νC-H, benzene), 1633 (νC=N, imine), 1608 (νC=N, benzothiazole), 1570, 1533, 1500 and 1444 (νC=C, benzene), 1363 (νN=N), 831 (δo.o.p.C-H, benzene).

2.2.3. **General procedure for the synthesis of imidazolidines 4a-h :**

A mixture of equimolar amounts of imine derivatives **3a-h** (1 mmol) and L-alanine (0.089 g, 1 mmol) in tetrahydrofuran (1 mL) was heated (550-610W) in microwave oven for 20-25 min. TLC (*n*-hexane: EtOAc) showed that the reactions were completed. The products were washed with diethyl ether and recrystallized from ethanol.

(E)-2-(5-(benzo[d]thiazol-2-yldiazenyl)-2- hydroxyphenyl)-5-methyl-3-(4-nitrophenyl)imidazolidin-4-one (4a):): IR (cm^{-1}): 3234_{br} (νO-H and νN-H, imidazolidine, vib. coupling), 3057 (νC-H, benzene), 1664 (νC=O and δN-H, imidazolidine, vib. coupling), 1597 (νC=N, benzothiazole), 1510 (νas.NO_2), 1452 (νC=C, benzene), 1421 (νN=N), 1286 (νs.NO_2), 754 (δo.o.p.C-H, benzene); ^{1}H NMR: δ (ppm) = 1.24 (d, *J* = 6.7 Hz, 3H, CH_3), 3.84 (q, *J* = 6.7 Hz, 1H, CH–CH_3, imidazolidine), 6.52 (d, *J* = 7.8 Hz, 1H, N–CH–N, imidazolidine), 6.67–7.61 (11H, Ar–H), 8.07 (s, 1H, N–H, imidazolidine), 9.51 (s, 1H, O–H). The singlet signals around 2.49 ppm and 3.42 ppm attributed to DMSO and absorbed H_2O in DMSO, respectively; Anal. Calcd. for $C_{23}H_{18}N_6O_4S$: C, 58.22; H, 3.82; N, 17.71; S, 6.76; Found C, 57.86; H, 3.49; N, 17.42; S, 6.67 . .

(E)-2-(5-(benzo[d]thiazol-2-yldiazenyl)-2 -hydroxyphenyl)-5-methyl-3-(3-nitrophenyl)imidazolidin -4-one (4b): IR (cm^{-1}): 3203_{br} (νO-H and νN-H, imidazolidine, vib. coupling), 3064 (νC-H, benzene), 1668 (νC=O, imidazolidine), 1653 (δN-H, imidazolidine), 1604 (νC=N, benzothiazole), 1558 and 1454 (νC=C, benzene), 1510 (νas.NO_2), 1423 (νN=N), 1288 (νs.NO_2), 825 (δo.o.p.C-H, benzene); ^{1}H NMR: δ (ppm) = 1.24 (d, *J* = 6.7 Hz, 3H, CH_3), 3.78 (q, *J* = 7.0 Hz, 1H, CH–CH_3, imidazolidine), 6.52 (d, *J* = 8.1 Hz, 1H, N–CH–N, imidazolidine), 6.66–7.61 (11H, Ar–H), 8.06 (s, 1H, N–H, imidazolidine), 9.49 (s, 1H, O–H). The singlet signals around 2.49 ppm and 3.42 ppm assigned to DMSO and absorbed H_2O in DMSO, respectively; Anal. Calcd. for $C_{23}H_{18}N_6O_4S$: C, 58.22; H, 3.82; N, 17.71; S, 6.76; Found C, 58.57; H, 3.52; N, 17.35; S, 6.62 . .

(E)-2-(5-(benzo[d]thiazol-2-yldiazenyl)-2- hydroxyphenyl)-3-(4-hydroxyphenyl)-5-methylimidazolidin -4-one (4c):

IR (cm^{-1}): 3209_{br} (νO-H and νN-H, imidazolidine, vib. coupling), 3093 (νC-H, benzene), 2987 (νas.C-H_3), 1668 (νC=O and δN-H, imidazolidine, vib. coupling), 1599 (νC=N, benzothiazole), 1514 and 1456 (νC=C, benzene), 1413 (νN=N), 833 (δo.o.p.C-H, benzene); ^{1}H NMR: δ (ppm) = 1.24 (d, *J* = 6.9 Hz, 3H, CH_3), 3.84 (q, *J* = 7.0 Hz, 1H, CH–CH_3, imidazolidine), 6.51 (d, *J* = 8.1 Hz, 1H, N–CH–N, imidazolidine), 6.66–7.66 (11H, Ar–H), 8.07 (s, 1H, N–H, imidazolidine), 10.12 (s, 2H, 2×O–H). The singlet signals around 2.49 ppm and 3.42 ppm due to DMSO and absorbed H_2O in DMSO, respectively; Anal. Calcd. for $C_{23}H_{19}N_5O_3S$: C, 62.01; H, 4.30; N, 15.72; S, 7.20; Found C, 61.61; H, 4.16; N, 15.59; S, 6.91.

(E)-2-(5-(benzo[d]thiazol-2-yldiazenyl)-2- hydroxyphenyl)-3-(4-methoxyphenyl)-5-methylimidazolidin-4-one (4d):

IR (cm^{-1}): 3203 (νO-H), 3171 (νN-H, imidazolidine), 3070 (νC-H, benzene), 2943 (νas.CH_3) 1674 (νC=O, imidazolidine), 1658 (δN-H, imidazolidine), 1606 (νC=N, benzothiazole), 1516 and 1454 (νC=C, benzene), 1427 (νN=N), 827 (δo.o.p.C-H, benzene); ^{1}H NMR: δ (ppm) = 1.23 (d, *J* = 6.7 Hz, 3H, CH_3), 3.61 (s, 1H, O–CH_3), 3.85 (q, *J* = 6.6 Hz, 1H, CH–CH_3, imidazolidine), 6.52 (d, *J* = 8.0 Hz, 1H, N–CH–N, imidazolidine), 6.74–7.60 (11H, Ar–H), 8.06 (s, 1H, N–H, imidazolidine), 9.51 (s, 1H, O–H). The singlet signals around 2.49 ppm and 3.34 ppm attributed to DMSO and absorbed H_2O in DMSO, respectively; Anal. Calcd. for

$C_{24}H_{21}N_5O_3S$: C, 62.73; H, 4.61; N, 15.24; S, 6.98; Found C, 62.44; H, 4.32; N, 15.21; S, 6.75

(E)-2-(5-(benzo[d]thiazol-2-yldiazenyl)-2- hydroxyphenyl)-3-(2-methoxyphenyl)-5-methylimidazolidin-4-one (4e): 3196_{br} (νO-H and νN-H, imidazolidine, vib. coupling), 3059 (νC-H, benzene), 1674 (νC=O, imidazolidine), 1660 (δN-H, imidazolidine), 1612 (νC=N, benzothiazole), 1548, 1512 and 1452 (νC=C, benzene), 1425 (νN=N), 821 (δo.o.p.C-H, benzene); ^{1}H NMR: δ (ppm) = 1.24 (d, *J* = 6.7 Hz, 3H, CH_3), 3.84 (q, *J* = 6.8 Hz, 1H, CH–CH_3, imidazolidine), 4.16 (s, 1H, O–CH_3), 6.53 (d, *J* = 8.1 Hz, 1H, N–CH–N, imidazolidine), 6.70–7.73 (11H, Ar–H), 8.06 (s, 1H, N–H, imidazolidine), 9.64 (s, 1H, O–H). The singlet signals around 2.49 ppm and 3.42 ppm assigned to DMSO and absorbed H_2O in DMSO, respectively; Anal. Calcd. for $C_{24}H_{21}N_5O_3S$: C, 62.73; H, 4.61; N, 15.24; S, 6.98; Found C, 62.36; H, 4.35; N, 15.15; S, 6.81 .
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(E)-2-(5-(benzo[d]thiazol-2-yldiazenyl)-2- hydroxyphenyl)-3-(4-bromophenyl)-5-methylimidazolidin-4-one (4f): IR (cm^{-1}): 3188_{br} (νO-H and νN-H, imidazolidine, vib. coupling), 3061 (νC-H, benzene), 2943 (νCH_3), 1668 (νC=O and δN-H, imidazolidine, vib. coupling), 1604 (νC=N, benzothiazole), 1504 and 1452 (νC=C, benzene), 1425 (νN=N), 823 (δo.o.p.C-H, benzene); ^{1}H NMR: δ (ppm) = 1.24 (d, *J* = 6.8 Hz, 3H, CH_3), 3.84 (q, *J* = 7.0 Hz, 1H, CH–CH_3, imidazolidine), 6.64 (d, *J* = 8.0 Hz, 1H, N–CH–N, imidazolidine), 7.00–7.61 (11H, Ar–H), 8.06 (s, 1H, N–H, imidazolidine), 9.50 (s, 1H, O–H). The singlet signals around 2.49 ppm and 3.42 ppm due to DMSO and absorbed H_2O in DMSO, respectively; Anal. Calcd. for $C_{23}H_{18}N_5O_2SBr$: C, 54.34; H, 3.57; N, 13.78; S, 6.31; Found C, 53.96; H, 3.33; N, 13.43; S, 6.09 .

(E)-2-(5-(benzo[d]thiazol-2-yldiazenyl)-2- hydroxyphenyl)-3-(4-chlorophenyl)-5-methylimidazolidin-4-one (4g): IR (cm^{-1}): 3419_{br} (νO-H and νN-H, imidazolidine, vib. coupling), 3063 (νC-H, benzene), 1680 (νC=O, imidazolidine) 1651 (δN-H, imidazolidine), 1618 (νC=N, benzothiazole), 1558, 1539, 1512 and 1454 (νC=C, benzene), 1421 (νN=N), 817 (δo.o.p.C-H, benzene); ^{1}H NMR: δ (ppm) = 1.24 (d, *J* = 6.7 Hz, 3H, CH_3), 3.84 (q, *J* = 6.8 Hz, 1H, CH–CH_3, imidazolidine), 6.59 (d, *J* = 7.5 Hz, 1H, N–CH–N, imidazolidine), 7.02–7.61 (11H, Ar–H), 8.06 (s, 1H, N–H, imidazolidine), 9.53 (s, 1H, O–H). The singlet signals around 2.49 ppm and 3.37 ppm attributed to DMSO and absorbed H_2O in DMSO, respectively; Anal. Calcd. for $C_{23}H_{18}N_5O_2SCl$: C, 59.54; H, 3.91; N, 15.10; S, 6.91; Found C, 58.19; H, 3.90; N, 15.01; S, 6.51.

(E)-2-(5-(benzo[d]thiazol-2-yldiazenyl)-2- hydroxyphenyl)-3-(2,4-dichlorophenyl)-5-methylimidazolidin-4-one (4h): IR (cm^{-1}): 3443_{br} (νO-H), 3213 (νN-H, imidazolidine), 3064 (νC-H, benzene), 1681 (νC=O, imidazolidine), 1651 (δN-H, imidazolidine, vib. coupling), 1610 (νC=N, benzothiazole), 1556, 1539, 1510 and 1456 (νC=C, benzene), 1423 (νN=N), 823 (δo.o.p.C-H, benzene); ^{1}H NMR: δ (ppm) = 1.23 (d, *J* = 6.8 Hz, 3H, CH_3), 3.84 (q, *J* = 7.0 Hz, 1H, CH–CH_3, imidazolidine), 6.59 (d, *J* = 7.1 Hz, 1H, N–CH–N, imidazolidine), 6.81–7.73 (10H, Ar–H), 8.06 (s, 1H, N–H, imidazolidine), 9.65 (s, 1H, O–H). The singlet signals around 2.49 ppm and 3.33 ppm due to DMSO and absorbed H_2O in DMSO, respectively; Anal. Calcd. for $C_{23}H_{17}N_5O_2SCl_2$: C, 55.43; H, 3.44; N, 14.05; S, 6.43; Found C, 54.06; H, 3.29; N, 13.94; S, 6.41.

2.3. Preliminary antibacterial assay

The antibacterial activities of the newly synthesized imidazolidines **4a-h** were determined by the agar diffusion method[29] using representative Gram (+) and Gram (–) bacteria on tryptic soya agar media. The test microorganisms to evaluate the potential antibacterial activity of the newly synthesized imidazolidines were *Staphylococcus aurous* (Gram-positive) and *Escherichia coli* (Gram-negative). The imidazolidines were dissolved in dimethylsulfoxide to prepare the test solutions of 10 mg/mL concentration. Gentamycin was used as a reference and the activities were presented as zones of inhibition for each compound (Table-2).

3. RESULTS AND DISCUSSION

3.1. Chemistry

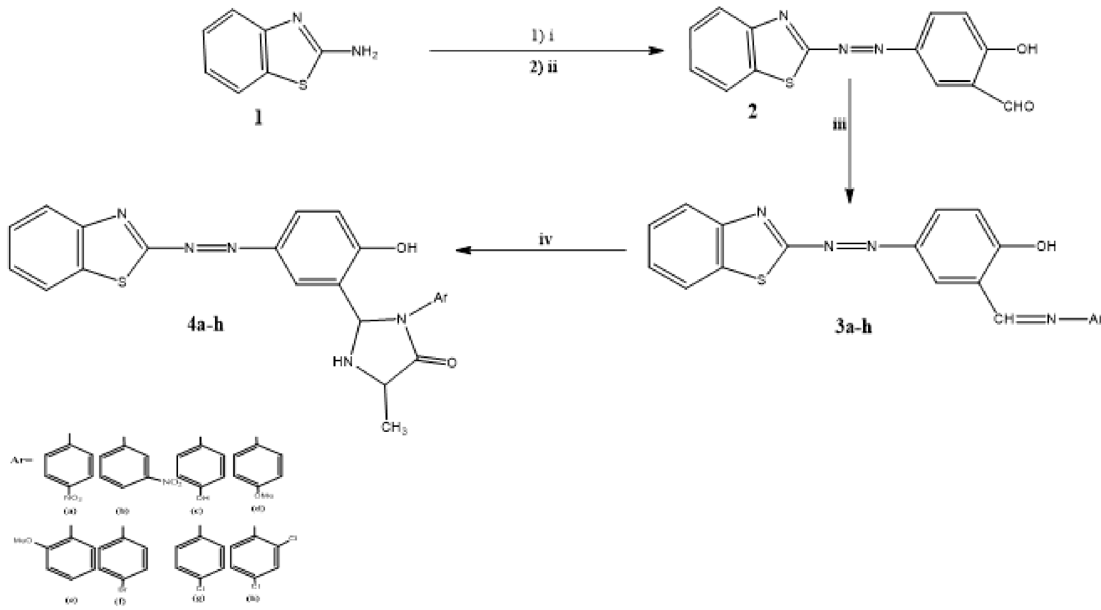
Amino group in 2-aminobenzothiazole **1** was diazotizated using sodium nitrite and sulforic acid to generate the corresponding diazonium salt which was directly introduced in coupling reaction with 2-hydroxybenzaldehyde dissolved in sodium hydroxide solution to give azo-benzothiazole derivative **2** containing aldehyde group. Aldehyde group of azo-benzothiazole derivative **2** was condensed with the primary aromatic amines including (4-nitroaniline, 3-nitroaniline, 4-hydroxyaniline, 4-methoxynitroaniline, 2-methoxyaniline, 4-bromoaniline, 4-chloroaniline and 2,4-dichloroaniline) using microwave irradiation in absolute ethanol to produce eight imine derivatives of benzothiazole **3a-h** respectively, as the platforms for this work (**Scheme-I** and **II**). The resulting imines **3a-h** were allowed to react with L-alanine using microwave irradiation leading to the formation of imidazolidine derivatives of benzothiazole **4a-h** in good yields (Table-1).

The chemical structures of the target compounds synthesized were deduced from IR, ^{1}H NMR spectral measurements and (CHNS) elemental analysis and were in good agreement with the proposed structures.

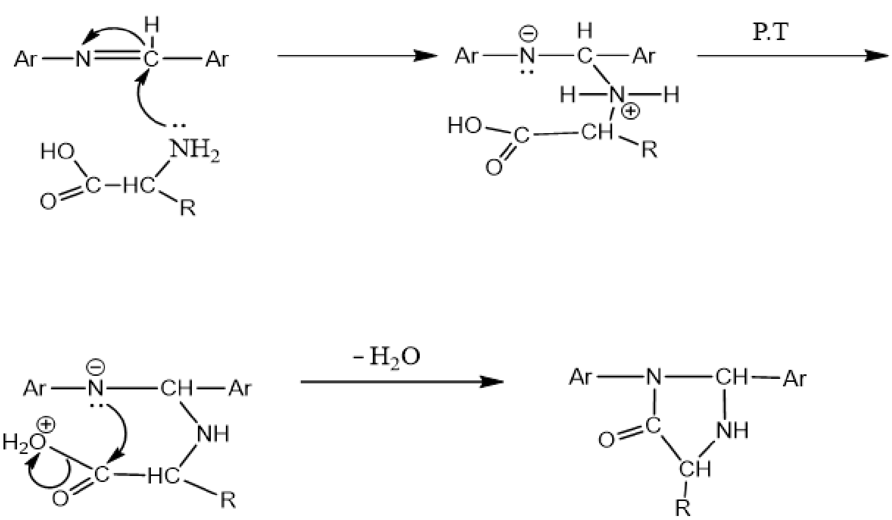
The IR and ^{1}H NMR spectra of the desired compounds **4a-h** were described in details in the Experimental section. The IR spectrum of azo-benzothiazole derivative **2** indicated the absence of a doublet band at 3375 and 3306 cm^{-1} for (NH_2)str in 2-aminobenzothiazole **1** and appearance of the following characteristic bands: the medium band at 1375 cm^{-1} attributed to azo group (N=N)str, the broad band at 3429 cm^{-1} assigned to (O-H)str, the sharp and strong band at 1645 cm^{-1} belong to aldehydic (C=O)str., the benzothiazolic (C=N)str appeared as strong band at 1610 cm^{-1}. IR spectra of the benzothiazolic-imines **3a-h** showed disappearing the sharp and strong band at 1645 cm^{-1} for aldehydic (C=O)str., also disappearing the sharp doublet band for (NH_2)str in the starting amines at the general range 3400-3250 cm^{-1} and appearing absorption band at the range 1604-1633 cm^{-1} assigned to the iminic (C=N)str. The benzothiazolic (C=N)str appeared at the range 1589-1610. The IR spectra of the benzothiazolic-imidazolidines **4a-h** showed the appearance of strong band at the range 1664-1710 cm^{-1} attributed to the (C=O)str of the imidazolidine ring. The spectra also showed the appearance of absorption band at the range 1651-1668 cm^{-1} assigned to the (N-H)bend of the imidazolidine ring. The benzothiazolic (C=N)str appeared at the range 1597-1618 cm^{-1}.

The structures of the imidazolidine compounds **4a-h** were confirmed by their ^1H NMR spectra which appeared doublet signal at δ 1.24, 1.24, 1.24, 1.23, 1.24, 1.24, 1.24 and 1.23 ppm, respectively belong to the methyl (CH_3) protons, the peak for the methine proton (C**H**-CH_3) of imidazolidine ring as a quartet at 3.84, 3.78, 3.84, 3.85, 3.84, 3.84, 3.84 and 3.84 ppm, respectively. The (N-CH-N) proton of imidazolidine appeared as a doublet at 6.52, 6.52, 6.51, 6.52, 6.53, 6.64, 6.59 and 6.59 ppm, respectively. The (Ar-H) protons at δ 6.66 –7.73 ppm, the (N-H) proton of imidazolidine ring as a singlet at 8.07, 8.06, 8.07, 8.06, 8.06, 8.06, 8.06 and 8.06 ppm, respectively. The (O-H) proton as a singlet at 9.51, 9.49, 10.12, 9.51, 9.64, 9.50, 9.53 and 9.65 ppm, respectively. The methoxy protons (O-CH_3) in compounds **4d** and **4e** appeared as a singlet at δ 3.61 and 4.16 ppm, respectively.

Moreover, the (CHNS) elemental analysis results were within ± 0.4% of the theoretical values and in good agreement with the proposed chemical structures for compounds **4a-h** given in the experimental section.



Scheme-I: Synthesis of imidazolidines, Reagents and conditions (i) $NaNO_2$, H_2SO_4, 0 ºC; (ii) 2-hydroxybenzaldehyde, NaOH 10% , 5ºC; (iii) Ar-NH_2, EtOH, MW (300-350W), 25-30 min; (iv) L-alanine, THF, MW (550-610W), 20-25 min.



Scheme-II: Proposed mechanism for the formation of imidazolidine ring

3.1. Antibacterial activities

The antibacterial activities of the newly synthesized imidazolidines **4a-h** were evaluated by the agar diffusion method[29] using representative standard strains of Gram (+) and Gram (–) bacteria on tryptic soya agar media, as listed in Table-2. Dimethylsulfoxide was used as solvent for the test compounds.

Imidazolidine compounds **4c**, **4d**, **4f**, **4g** and **4h** were found to be greater activity than gentamycin against Gram-positive bacteria. Moreover, compounds **4c** and **4f** were also showed better activity to the control drug against Gram-negative bacteria.

TABLE-1: physical properties of the synthesized compounds

Product	Physical state	R_f (developer)	Mp (ºC)	Yield (%)	Time(min)	MW (W)
2	Dark brown solid	0.68 (*n*-hexane/ EtOAc, 1:2)	141-143	60	-	-
3a	Brown solid	0.78 (*n*-hexane/ EtOAc, 1:2)	221-223	78 350	30	350
3b	Dark brown solid	0.73 (*n*-hexane/ EtOAc, 1:2)	184-186	75 320	28	320
3c	Dark brown solid	0.75 (*n*-hexane/ EtOAc, 1:2)	176-178	79 300	27	300
3d	Brown solid	0.88 (*n*-hexane/ EOAc, 1:2)	195-197	73 300	26	300
3e	Dark brown solid	0.70 (*n*-hexane/ EtOAc, 1:2)	199-201	77 310	30	310
3f	Dark Brown solid	0.83 (*n*-hexane/ EtOAc, 1:2)	173-175	69 320	30	320
3g	Brown solid	0.75 (*n*-hexane/ EtOAc, 1:2)	206-208	76 320	28	320
3h	Brown solid	0.70 (*n*-hexane/ EtOAc, 1: 2)	163-165	77 310	30	310
4a	Yellow solid	0.69 (*n*-hexane/ EtOAc, 1: 2)	247-249	74 610	24	610
4b	Dark orange solid	0.71 (*n*-hexane/ EtOAc, 1: 2)	241-243	73 600	23	600
4c	Orange solid Dark	0.71 (*n*-hexane/ EtOAc, 1: 2)	297-299	70 600	22	600
4d	Orange solid	0.69 (*n*-hexane/ EtOAc, 1: 2)	232-234	74 560	21	560
4e	Orange solid	0.68 (*n*-hexane/ EtOAc, 1: 2)	202-204	77 550	20	550
4f	Dark orange solid	0.73 (*n*-hexane/ EtOAc, 1: 2)	279-281	76 570	24	570
4g	Yellow solid	0.74 (n-hexane/ EtOAc , 1:2)	318(dec.)	78 600	22	600
4h	Dark brown solid	0.72 (*n*-hexane/ EtOAc, 1: 2)	286-288	79 600	22	600

TABLE-2 the antibacterial activity of compounds **4a-h** and gentamycin as control drug

Product	*Staphylococcus aureus* (Gram-positive)	*Escherichia coli* (Gram-negative)
4a	0	0
4b	11	0
4c	20	17
4d	19	0
4e	12	0
4f	20	16
4g	18	0
4h	22	0
DMSO	0	0
Gentamycin	15	15

The microwave irradiation is efficient technique including short reaction time and high yield in comparison with the classical thermal method. The rates of reactions of imines with alanine for formation of imidazolidines are approximately equal. All synthesized imidazolidines could be converted to the corresponding phenol salts which are completely soluble in water. The synthesized imidazolidines appeared higher biological action against Gram-positive bacteria than that of Gram-negative bacteria. Imidazolidine compounds **4c**, **4d**, **4f**, **4g** and **4h** were found to be greater activity than gentamycin against Gram-positive bacteria, while compounds, **4c** and **4f** showed better activity to the control drug against Gram-negative bacteria.

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